

TessScope

Exact cross-runtime gradients for closed-loop microscope co-design

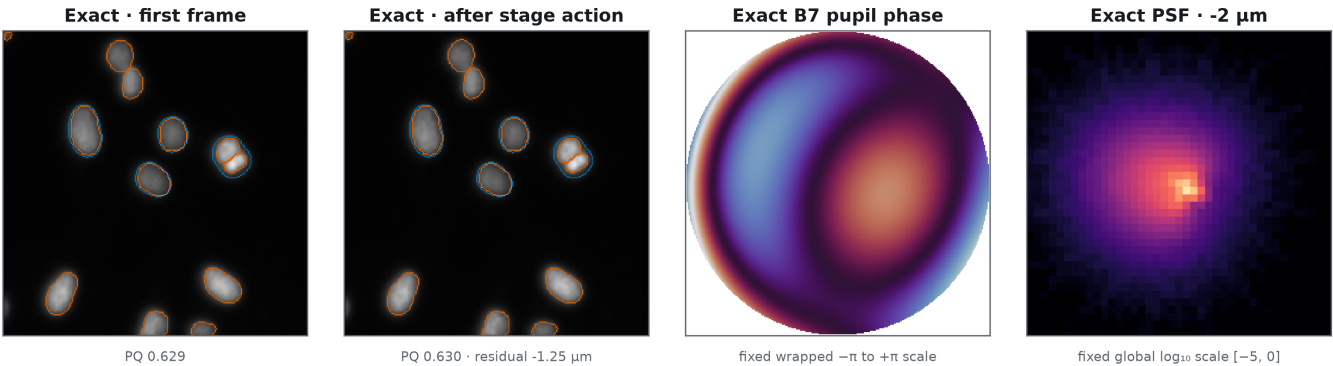
Tesseract Hackathon 2026 – Track 05 – Differentiable graphics & rendering

The problem. A fluorescence microscope pupil that preserves nucleus segmentation tends to erase depth cues, while a pupil that makes signed defocus obvious can distort nuclei. TessScope co-designs one bounded phase-only pupil for the complete two-exposure decision loop rather than optimizing image sharpness in isolation.

Protected validation claim. TessScope uses exact gradients across JAX/Chromatix optics, NumPy/SciPy autofocus, and PyTorch/InstanSeg to causally improve closed-loop correction over a forward-identical stopped-gradient system.

One frozen TessScope validation field, from optics to correction

n21_s1 · initial defocus -2 μm · exact-gradient design · deterministic cached evidence



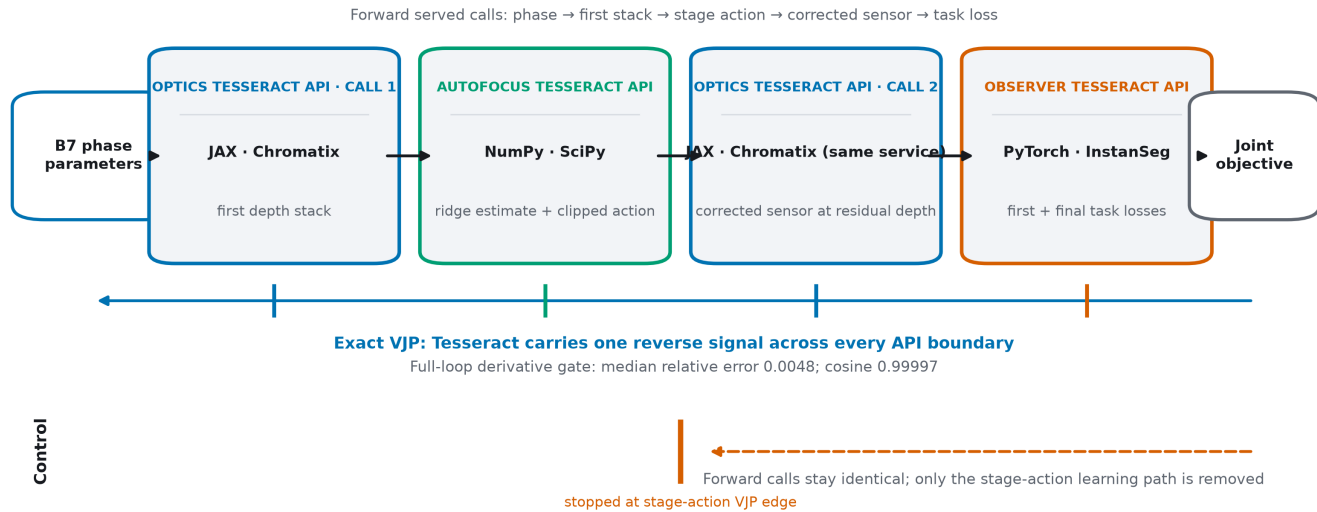
Blue: automated reference · orange: InstanSeg · magenta: overlap · representative frame only; population result reported separately

Measure	Result	Meaning
First hard off-focus PQ	0.492192	Before the predicted stage move
Corrected hard off-focus PQ	0.552295	One bounded predicted correction later
Focus error	1.267923 μm	98.89% signed-direction accuracy
Exact vs. stopped	+0.006259 PQ	95% well-bootstrap CI $[+0.000670, +0.011547]$
Exact vs. piecewise-028	+0.001285 PQ	CI $[-0.001719, +0.004800]$; not significant

Scope. PQ is panoptic quality on a 0–1 scale, not percent accuracy. All displayed v2 evidence is validation-only. The locked BBBC006 test remains sealed. No physical microscope or fabricated phase mask has validated the system.

One differentiable program, three native runtimes

Tesseract bridges the exact gradient across isolated scientific runtimes



Tesseract	Native stack	Forward role	Reverse path
Optics	JAX + Chromatix	Phase pupil and depth-dependent fluorescence formation; called for both exposures	JAX VJP through pupil and residual depth
Autofocus	NumPy + SciPy	Spectral features, signed-z ridge fit, clipped stage action	Analytic feature VJP and implicit ridge-solve VJP
Observer	PyTorch + InstanSeg	Frozen nucleus task loss on first and corrected frames	PyTorch autograd input VJP

Why Tesseract is load-bearing. Each component stays in its tested scientific ecosystem. Tesseract supplies typed service boundaries and derivative endpoints; `tesseract-jax` exposes remote calls as a JAX-differentiable program. One objective therefore returns one pupil gradient through JAX → SciPy → JAX → PyTorch without reimplementing the models in a single framework.

Closed-loop computation.

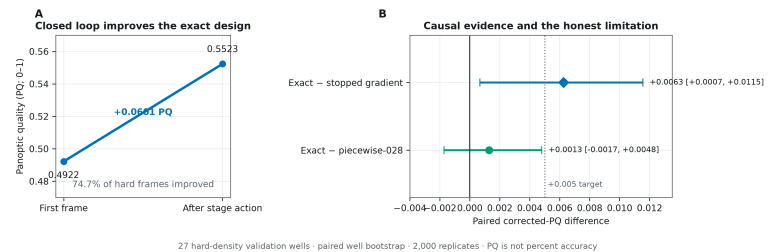
1. The bounded seven-parameter pupil forms an initial seven-plane stack.
2. Two support fields fit a signed-depth autofocus model; the query predicts a stage move clipped to $[-6, +6]$ μm .
3. The move changes residual depth and the same pupil forms a corrected exposure.
4. Frozen InstanSeg losses score both frames; exact VJPs optimize the full loop.

Derivative evidence. Five-direction central differences passed with median relative error 0.004802 and cosine agreement 0.999967. Exact and stopped modes differed only in the action's reverse path and had measured forward difference 0.0. The source composition is `src/tessscope/v2_3/closed_loop.py`; all three API definitions are under `services/`.

The ablation is causal: identical starts, forward values, objective profiles, budgets, and schedules; only the gradient through the autofocus action and downstream residual-depth terms is stopped.

Frozen protocol and paired evidence

Exact gradients improve closed-loop correction over a stopped-gradient control



Dataset and split

- BBBC006v1 U2OS Hoechst fluorescence.
- z13–z19: –6 to +6 μm around z16 in 2 μm steps.
- Well-grouped deterministic split: 288 train, 48 validation, 48 sealed test wells.
- Same 45 registration-valid validation wells for every frozen design; 27 preregistered hard-density wells are the primary comparison unit.

Uncertainty and selection

- 2,000 deterministic paired bootstrap replicates re-sample wells, not frames.
- 270 existing v2.3 checkpoints were audited without retraining.
- Three were frozen before hard-label evaluation by a preregistered one-per-run rule.
- Reference instances are automated CellProfiler-derived labels, not manual truth.

What the result supports. The balanced exact checkpoint raised corrected PQ by 0.006259 over its matched stopped-gradient pupil, with a positive paired 95% interval. Its own first-to-corrected PQ rose from 0.492192 to 0.552295, and 74.7% of hard frames improved after one predicted action.

What it does not support. Exact optimization beat piecewise-028 by only 0.001285 PQ; the interval crosses zero and only 55.6% of hard wells favor exact, below the frozen 60% stability rule. First-frame PQ also missed the segmentation-only tolerance by 0.000295. Therefore no derivative-free or locked-test branch was authorized.

Phase	Question	Outcome
v2.5	Can exact constrained B11/B7 optimization resolve the near miss?	No endpoint passed the frozen training and soft-validation constraints.
v2.6	Can controller/exposure or sequential masks create enough headroom?	Diagnostics and the complete two-mask matrix produced zero promotable candidates.

Negative follow-ups are complete experimental outcomes. They are preserved in docs/RESEARCH_HISTORY.md; they are not presented as positive results or hidden behind the demo.

Reproducibility, track fit, and limitations

Two reproducibility levels.

Path	Command and inputs	Purpose
Instant judge replay	<code>python3 scripts/serve_demo.py</code>	Standard library only; no data, model, training, or live inference
Full scientific path	Python 3.12, <code>uv sync -frozen</code> , official hashed assets, three local Tesseract	Re-run derivative, optimization, and 45-well validation into isolated checkpoints

The repository includes a 12.4 MiB validation replay, frozen claim matrix, JSON-pointer traceability, SHA-256 manifests, deterministic figure builders, a committed lock file, and output-isolation options. The full instructions are in `docs/FULL_REPRODUCTION.md`. The approximately 2.5 GB source subset, model weights, virtual environment, caches, and runtime checkpoints are not committed.

Track 05 fit. TessScope’s central artifact is differentiable image formation: the optimized phase pupil changes a rendered fluorescence observation, which changes a served focus decision, which changes the next rendered observation and biological task loss. Tesseract is essential because the end-to-end derivative crosses three runtime and framework boundaries with independently tested VJPs.

Limitations.

- In-silico digital-twin evidence cannot establish laboratory performance, calibration robustness, SLM efficiency, bleaching, or motion tolerance.
- The supported causal effect is validation-only and is smaller than the preregistered improvement required over the strongest piecewise baseline.
- BBBC006 reference masks are automated and may contain systematic errors.
- The cached demo is an evidence viewer; it does not imply a production service.

Primary sources and licenses.

- Pasteur Labs, *Tesseract Hackathon 2026* and Tesseract differentiable-pipeline documentation.
- Ljosa et al., *Nature Methods* (2012), BBBC006v1; Broad Bioimage Benchmark Collection.
- Deb et al., *Chromatix*, bioRxiv (2025), DOI 10.1101/2025.04.29.651152.
- Goldsborough et al., *InstanSeg*, arXiv (2024), DOI 10.48550/arXiv.2408.15954.

TessScope code and original documentation are Apache-2.0. External datasets, models, software, fonts, and trademarks retain their own terms; see `LICENSE`, `NOTICE`, and `THIRD_PARTY_NOTICES.md`.

Submission boundary: local release candidate only. No repository push, deployment, publication, video upload, hackathon submission, or locked-test access is performed by this package.